॥ सा विद्या या विमुक्तये ॥
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CS101C: Introduction to Programming (Using C)

Instructors: Prof. Aikta Arya and Prof. Konjengbam Anand

CSE Department

Indian Institute of Technology, Dharwad

Week 7

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Today's Class (15/9/2026)
Practice session 2

String MCQ 1 — The Basics!

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What is printed?

```
char s[] = "HELLO";

printf("%d", sizeof(s));
```

A. 5

B. 6

C. 4

D. Depends on compiler

Answer — MCQ 1

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B. 6

H E L L O \0

Important:

"HELLO" $= 5$ *characters* $+ 1$ *null character*

String MCQ 2 — Array Size Trap

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What is the value of `strlen(s)`?

```
char s[] = {'C','S','E','\0'};

printf("%d", strlen(s));
```

A. 3

B. 4

C. 2

D. Error

Answer — MCQ 2

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A. 3

`C S E \0`

`strlen()` counts characters before `\0`.

`strlen(s) = 3`

String MCQ 3 — Can You See the Problem?

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What is wrong with this code?

```
char s[5] = "HELLO";

printf("%s", s);
```

A. Nothing is wrong
B. Array is too small for the string
C. `printf` cannot print strings
D. `HELLO` is not a string

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B. Array is too small

$$\texttt{HELLO} = 5\ characters$$

but a C string also needs:

$$\backslash 0$$

Therefore:

```
char s[6] = "HELLO";
```

is required.

No room for the null terminator!

String MCQ 4 — Indexing Trap

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```
char s[] = "COMPUTER";

printf("%c", s[4]);
```

A. P

B. U

C. T

D. E

Answer — MCQ 4

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$index$	0	1	2	3	4	5	6	7
$char$	C	O	M	P	U	T	E	R

B. U

String MCQ 5 (a) — Modification!

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What is printed?

```
char s[] = "HELLO";

s[1] = 'A';

printf("%s", s);
```

A. HELLO

B. HALLO

C. HAALO

D. Error

Answer — MCQ 5

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Initially:

```
H E L L O
```

Then:

```
s[1] = 'A'
```

Therefore:

HALLO

Character arrays can be modified!

String MCQ 5 (b) — Modification!

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What is printed?

```
char *s = "HELLO";

s[1] = 'A';

printf("%s", s);
```

A. undefined behavior

B. HALLO

C. HAALO

D. Error

```
char s[] = "HELLO";
char *p2 = s;
p2[0] = 'Y'; // OK
s[1] = 'X'; // OK
```



Character array — MODIFIABLE

VS

```
char *s = "HELLO";
char *p2 = s;
p2[0] = 'Y'; // DON'T
s[1] = 'X'; // DON'T
```



String literal — DON'T MODIFY

Golden Rule:

A pointer can modify characters **only when it points to modifiable memory.**

String MCQ 6 — The \0 Trap

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What will `printf("%s",s)` print?

```
char s[] = {'H','I','\0','X','Y'};

printf("%s", s);
```

A. HIXY

B. HI

C. HIX

D. Error

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H I \0 X Y

B. HI

printf("%s",s) stops at the first:

\0

String MCQ 7 — sizeof vs strlen

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```
char s[] = "DATA";

printf("%d %d", sizeof(s), strlen(s));
```

A. 4 4

B. 5 4

C. 4 5

D. 5 5

Answer — MCQ 7

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D A T A $\backslash 0$
Therefore:

`sizeof(s) = 5`

but

strlen(s) $= 4$

B. 5 4

String MCQ 8 — Pointer + String

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```
char s[] = "HELLO";
char *p = s;

printf("%c", *(p + 2));
```

A. H

B. E

C. L

D. O

Answer — MCQ 8

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$$p \rightarrow s[0]$$

Therefore:

$$p + 2 \rightarrow s[2]$$

and:

$$*(p + 2) = s[2] = \texttt{'L'}$$

C. L

String MCQ 9 — Pointer Movement

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```
char s[] = "ABCDE";
char *p = s;

printf("%c ", *p);
p++;
printf("%c ", *p);
p += 2;
printf("%c", *p);
```

A. A B C

B. A B D

C. B C E

D. A C D

Answer — MCQ 9

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$$p = s[0]$$

First:

$$*p = \texttt{A}$$

After p++:

$$p = s[1] \Rightarrow \texttt{B}$$

After p+=2:

$$p = s[3] \Rightarrow \texttt{D}$$

B. A B D

String MCQ 10 — Reverse Thinking

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```
char s[] = "ABCDE";

s[0] = s[4];
s[4] = 'X';

printf("%s", s);
```

A. ABCDX

B. EBCDX

C. XBCDE

D. EBCEX

Answer — MCQ 10

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Initially:

```
A B C D E
```

After:

```
s[0]=s[4]
```

we get:

```
E B C D E
```

Then:

```
s[4]='X'
```

Therefore:

EBCDX

String MCQ 11 — VERY TOUGH

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What is printed?

```
char s[] = "HELLO";
char *p = s;

printf("%c ", *p++);
printf("%c ", *p);
printf("%c", *(p+2));
```

A. H E L

B. H E O

C. E E L

D. H L O

Answer — MCQ 11

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Remember:

$$\boxed{*p++ = *(p++)}$$

So first:

$$*p++ \rightarrow \mathtt{H}$$

then p moves to E.

$$*p \rightarrow \mathtt{E}$$

Finally:

$$*(p+2) = \mathtt{L}$$

A. H E L

String MCQ 12 — Find the Bug

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```
char s[5];

s[0]='H';
s[1]='E';
s[2]='L';
s[3]='L';
s[4]='O';

printf("%s", s);
```

What is the problem?

A. Nothing

B. `printf` cannot print character arrays

C. Missing `\0`

D. Array must contain 10 elements

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The array contains:

H E L L O

but:

Where is \0\0\0\0?

There is no null terminator.
Therefore it is **not a valid C string**.

C. Missing \0

String MCQ 13 — strlen Trap

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```
char s[] = "HELLO\0WORLD";

printf("%d", strlen(s));
```

A. 5

B. 10

C. 11

D. 12

Answer — MCQ 13

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Memory contains:
H E L L O $\backslash 0 WORLD \backslash 0$
But strlen() stops at the first null character.
Therefore:

A. 5

First \0 = STOP!

String MCQ 14 — Array Modification

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```
char s[] = "COMPUTER";

s[2] = '\0';

printf("%s", s);
```

A. COMPUTER

B. CO

C. COM

D. COTER

Answer — MCQ 14

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After:

`s[2]='\0'`

the array becomes:
`C O \0` $P\ U\ T\ E\ R$
`%s stops at \0.`

B. CO

FINAL STRING TRAP

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What is printed?

```
char s[] = "ABCDE";
char *p = s;

*p = 'X';
p += 2;
*p = 'Y';

printf("%s", s);
```

A. XBCYE

B. XBYDE

C. ABCYE

D. XBCDE

Answer — FINAL STRING TRAP

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Initially:

```
A B C D E
```

```
*p = 'X'
```

$\Rightarrow$ `X B C D E`

Then:

$$p+ = 2$$

so p points to ‘C‘.

```
*p = 'Y'
```

$\Rightarrow$ `X B Y D E`

B. XBYDE

C String Functions — Must Know!

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Function	Purpose	Example
`strlen()`	Find length	`strlen("Hello")` $\rightarrow 5$
`strcpy()`	Copy string	`strcpy(a,b)`
`strcat()`	Join strings	`strcat(a,b)`
`strcmp()`	Compare strings	`strcmp(a,b)`
`strchr()`	Find character	`strchr(s,'a')`
`strstr()`	Find substring	`strstr(s,"cat")`

Most of these are in `<string.h>`

strlen()- How Long Is the String?

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Syntax

```
strlen(string)
```

Example

```
#include <stdio.h>
#include <string.h>
int main()
{
    char s[] = "HELLO";

    printf("%d", strlen(s));
}
```

5

Remember:

`strlen() DOES NOT COUNT \0`

strcpy() — Copy One String

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Syntax

```
strcpy(destination, source);
```

Example

```
char source[] = "HELLO";
char dest[20];
strcpy(dest, source);
printf("%s", dest);
```

HELLO

dest ← source

strcat() — Join Two Strings

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Syntax

```
strcat(destination, source);
```

Example

```
char a[30] = "Hello ";
char b[] = "World";
strcat(a, b);
printf("%s", a);
```

`Hello World`

Important:

`a = a + b`

The destination array must have enough space.

strcmp() — Compare Strings

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Syntax

```
strcmp(s1, s2)
```

- Returns **0** $\rightarrow$ strings are equal
- Returns < 0 $\rightarrow$ first string is smaller
- Returns > 0 $\rightarrow$ first string is larger

Example

```
char a[] = "apple";
char b[] = "apple";
if(strcmp(a,b) == 0)
    printf("Same");
```

Same

STRING TRAP — == vs strcmp()

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Which is the correct way?

```
char a[] = "hello";
char b[] = "hello";
```

A.

```
if(a == b)
```

B.

```
if(strcmp(a,b) == 0)
```

STRING TRAP — == vs strcmp()

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Which is the correct way?

```
char a[] = "hello";
char b[] = "hello";
```

A.

```
if(a == b)
```

B.

```
if(strcmp(a,b) == 0)
```

B

For string contents, use strcmp().

strchr() — Find a Character

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Syntax

```
strchr(string, character)
```

Example

```
char s[] = "COMPUTER";

char *p = strchr(s, 'P');

if(p != NULL)
    printf("Found");
```

```
Found
```

Returns a pointer to the **first occurrence**.

strstr() — Find a Substring

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Syntax

```
strstr(string, substring)
```

Example

```
char s[] = "I love programming";
if(strstr(s, "love") != NULL)
    printf("Found");
```

```
Found
```

Remember:

`strchr()` $\rightarrow$ *character*

`strstr()` $\rightarrow$ *substring*

Common String Functions

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Function	Purpose
`strrev(s)`	Reverse string
`strupr(s)`	Convert to uppercase
`strlwr(s)`	Convert to lowercase

Example

`strrev("HELLO")` → `"OLLEH"`

`strupr("hello")` → `"HELLO"`

`strlwr("HELLO")` → `"hello"`

Note:

`strrev()`, `strupr()`, and `strlwr()` are **not part of standard ISO C**.

Their availability depends on the compiler/library.

Character Functions — ctype.h

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Function	Checks / Converts
`isupper(c)`	Is uppercase?
`islower(c)`	Is lowercase?
`isdigit(c)`	Is digit?
`isalpha(c)`	Is alphabet?
`isalnum(c)`	Is alphabet or digit?
`isspace(c)`	Is whitespace?
`toupper(c)`	Convert to uppercase
`tolower(c)`	Convert to lowercase

These functions belong to ctype.h

toupper() & tolower()

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Example

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char c = 'a';

    printf("%c\n", toupper(c));
    printf("%c\n", tolower('Z'));
}
```

toupper() & tolower()

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Example

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char c = 'a';

    printf("%c\n", toupper(c));
    printf("%c\n", tolower('Z'));
}
```

`toupper('a')` $\rightarrow$ `'A'`

`tolower('Z')` $\rightarrow$ `'z'`

isupper(), islower(), isdigit()

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Example

```
char c = '7';

if(isdigit(c))
    printf("It is a digit");

if(isupper('A'))
    printf("Uppercase");

if(islower('a'))
    printf("Lowercase");
```

isupper(), islower(), isdigit()

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Example

```
char c = '7';

if(isdigit(c))
    printf("It is a digit");

if(isupper('A'))
    printf("Uppercase");

if(islower('a'))
    printf("Lowercase");
```

`isdigit('7')` $\rightarrow$ true

`isupper('A')` $\rightarrow$ true

`islower('a')` $\rightarrow$ true

String Challenge: Reverse Every Word

Problem

Write a C program to **reverse every individual word** in a sentence, while keeping the **word order unchanged**.

Input

```
I love programming in C
```

Expected Output

```
I evol gnimmargorp ni C
```

Think Before Coding

Step 1: Find a word

`love`

↓

Step 2: Reverse it

`evol`

Step 3: Continue with the next word

`programming` → `gnimmargorp`

Step 4: Keep the spaces!

Restriction

With and without using `strrev()`.

String Challenge: Remove Consecutive Duplicates



Problem

Write a C program that removes **consecutive repeated characters** from a string.

Example 1

`aaabbccccccdaa`

$\Downarrow$

`abcda`

Example 2

`aabbccaabb`

$\Downarrow$

`abcab`

Important!

We remove **only consecutive duplicates**. `aabbccaabb` $\rightarrow$ `abcab`

Think!

If the current character is the same as the previous character...

Should we copy it to the output?

String Challenge: Toggle Case



Problem

Write a C program that converts:

UPPERCASE $\leftrightarrow$ **lowercase**

That is:

- Uppercase letter $\rightarrow$ Lowercase
- Lowercase letter $\rightarrow$ Uppercase
- Digits and special characters $\rightarrow$ unchanged

Input

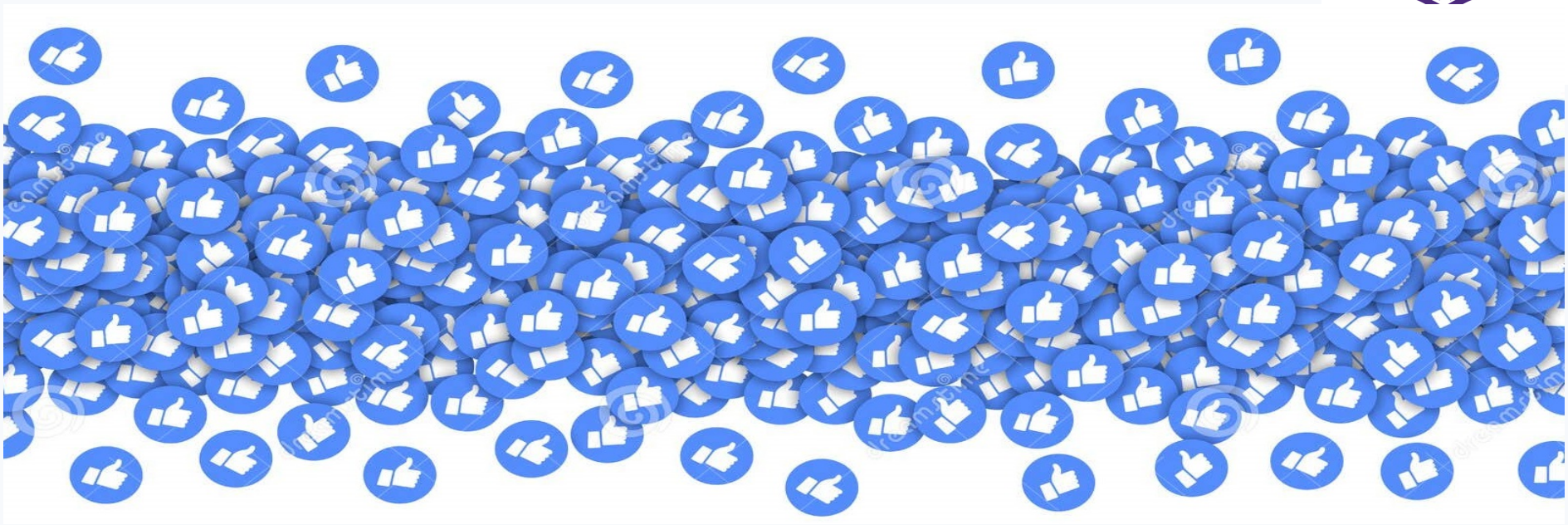
```
HeLLo WoRLd 123!
```

Output

```
hEllO wOrLd 123!
```

Restriction

with and without using `strupr()` or `strlwr()`.



Thank you for your kind attention!!